GREEEN tallgrass prairie restoration: Seeded richness ~ Seeding time * Herbicide pre-treatment (NW Station)

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> **NOTE:** To compare different models, you only have to change the models in the section 'Load models'

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```r
library(here)
library(tidyverse)
library(ggbeeswarm)
library(patchwork)
library(lme4)
library(glmmTMB)
library(DHARMa)
library(emmeans)
```

Packages

```r
sites <- read_csv(
  here("data", "processed", "data_processed_sites.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    id_plot = "f",
    year = "f",
    seeding_time = col_factor(
      levels = c("unseeded", "fall", "spring"), ordered = FALSE
      ),
    herbicide = col_factor(levels = c("0", "1"), ordered = FALSE),
    treatment_id = "f",
    richness_type = "f"
  )
) %>%
  filter(
    site == "NW Station",
    richness_type == "seeded_richness",
    treatment_id %in% c("1", "2", "3", "4")
  ) %>%
  mutate(
    treatment = str_c(seeding_time, herbicide, seeded_pool, sep = "_"),
    treatment = factor(treatment)
  ) %>%
  mutate(y = richness_1qm + richness_25qm)
```

Load data

Statistics

Data exploration

Means and deviations

```r
Rmisc::CI(sites$y, ci = .95)
```

```
##    upper     mean    lower
## 4.589483 4.270833 3.952184
```

```r
median(sites$y)
```

```
## [1] 4
```

```r
sd(sites$y)
```

```
## [1] 1.572656
```

```r
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##  5% 95%
##   2   7
```

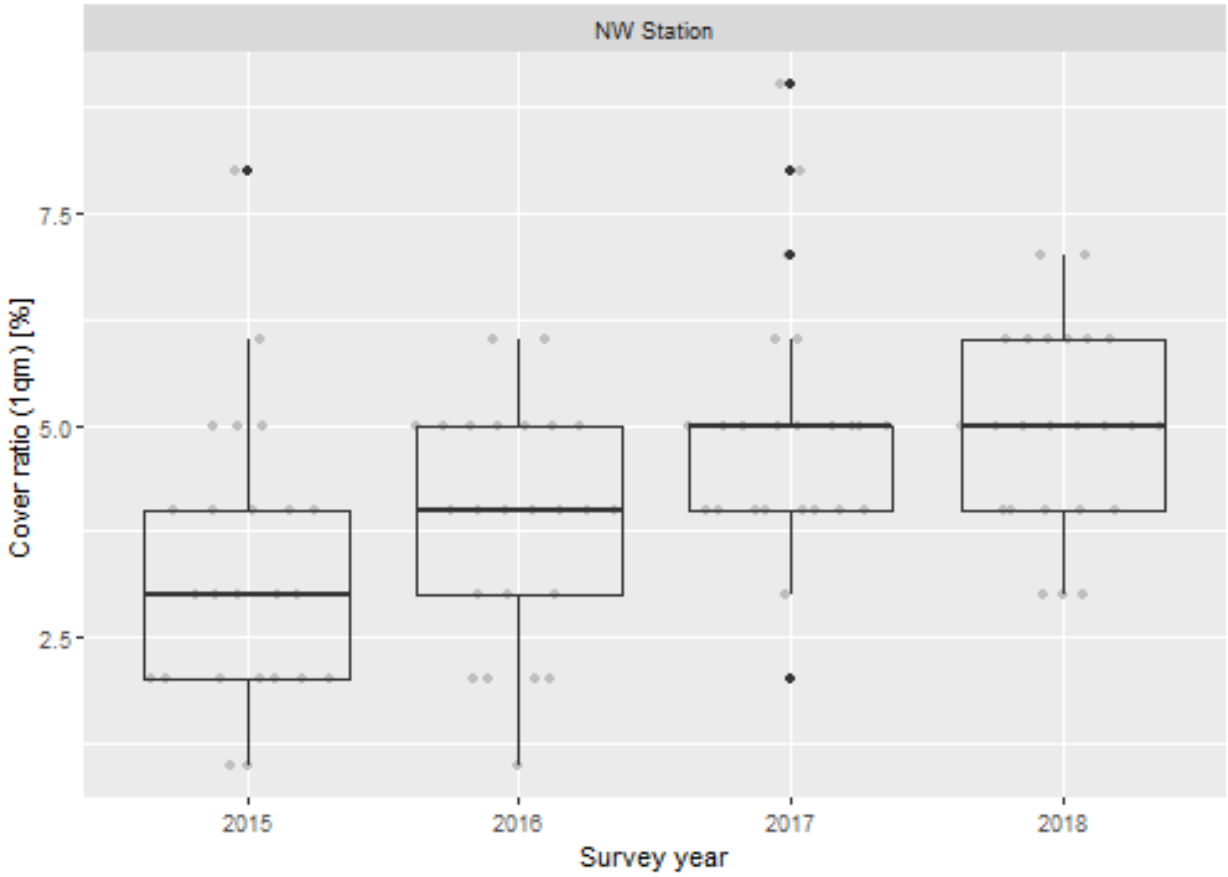
```r
sites %>% count(site, year)
```

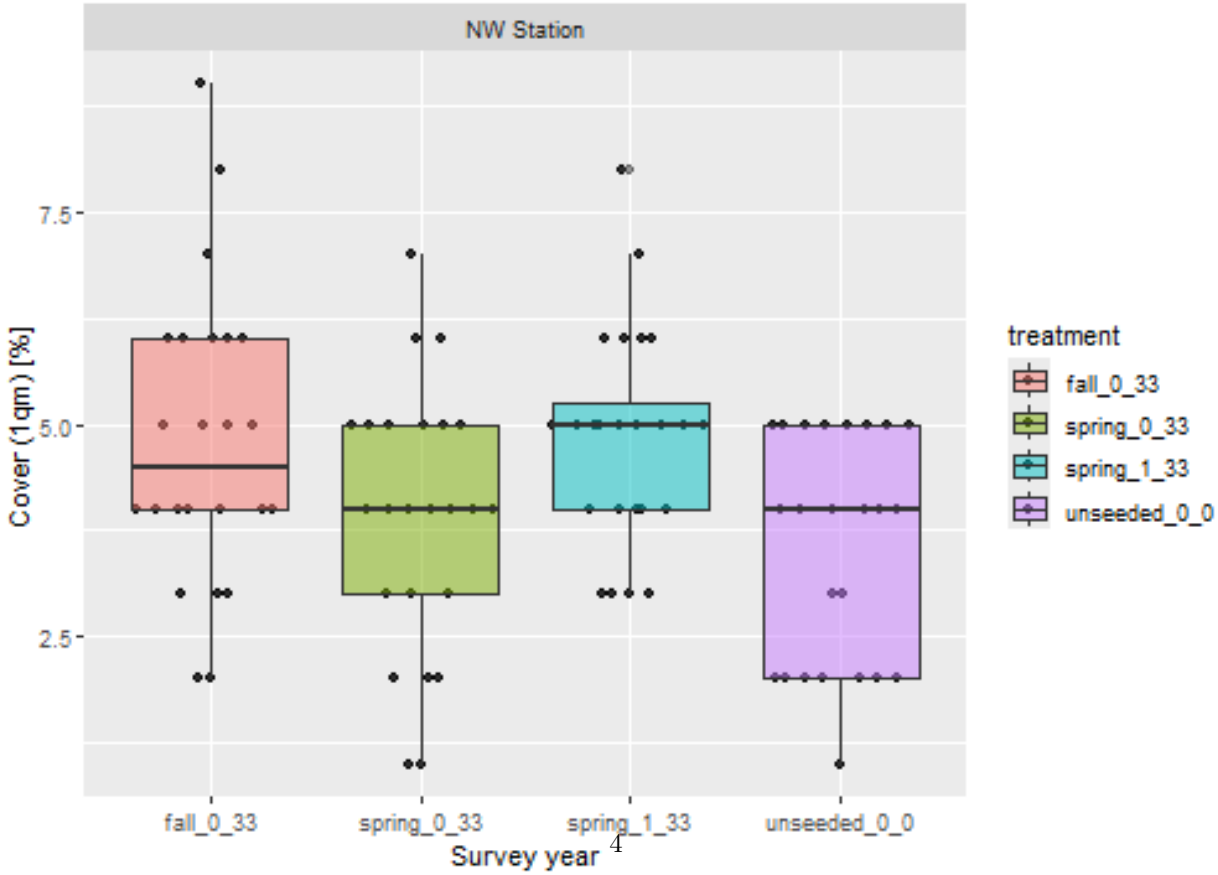
```
## # A tibble: 4 x 3
##   site       year      n
##   <chr>      <fct> <int>
## 1 NW Station 2015     24
## 2 NW Station 2016     24
## 3 NW Station 2017     24
## 4 NW Station 2018     24
```

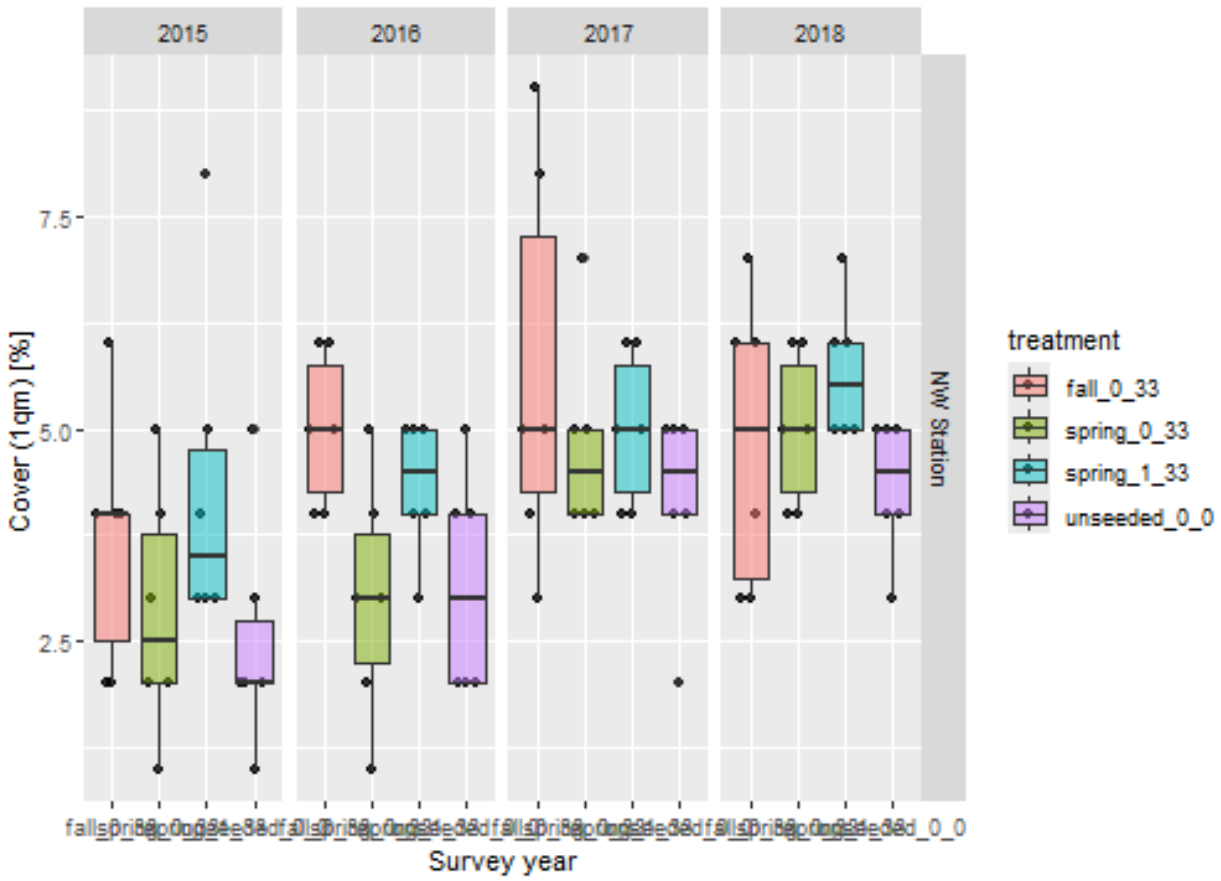
```r
sites %>% count(treatment)
```

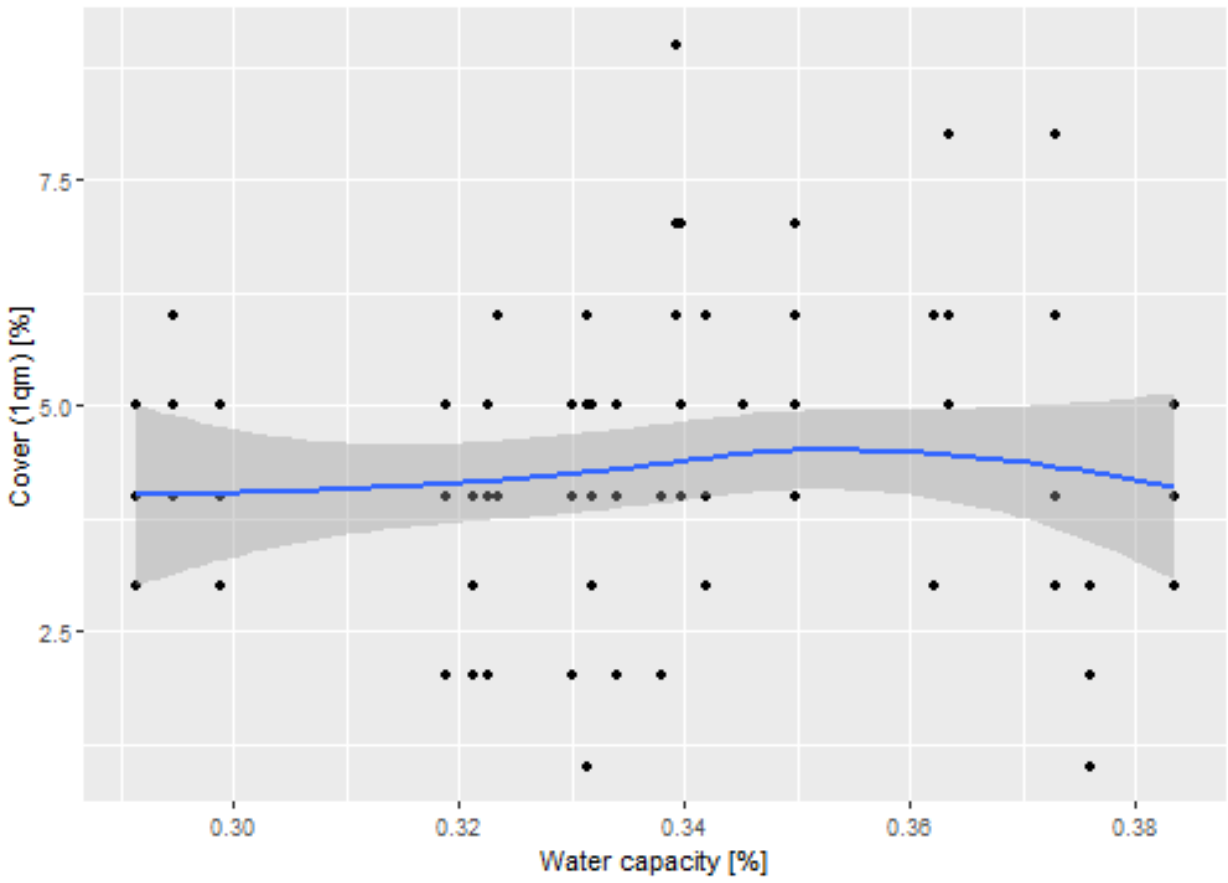
```
## # A tibble: 4 x 2
##   treatment        n
##   <fct>        <int>
## 1 fall_0_33       24
## 2 spring_0_33     24
## 3 spring_1_33     24
## 4 unseeded_0_0    24
```

Graphs of raw data (Step 2, 6, 7)

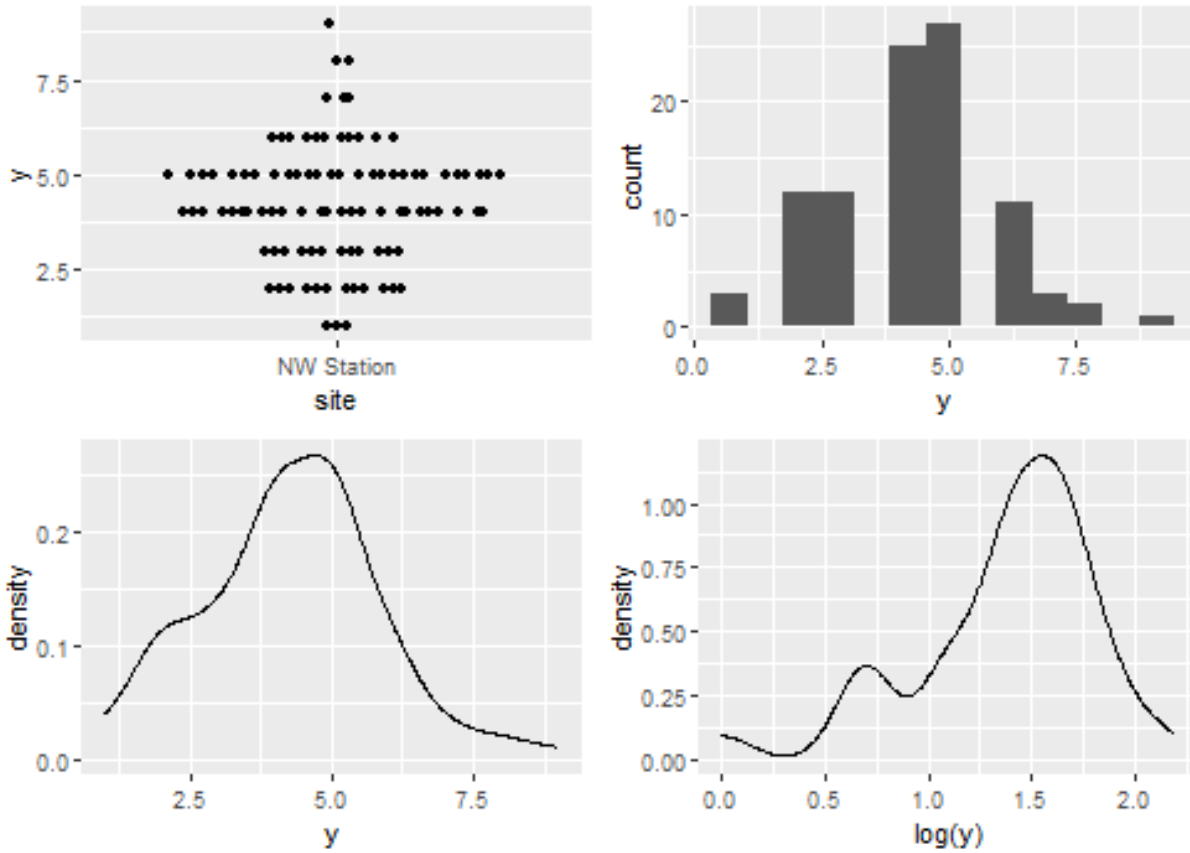





2015
2016
2017
2018
NW Station
Cover (1qm) [%]
7.5
5.0
2.5
Survey year
treatment
fall_0_33
spring_0_33
spring_1_33
unseeded_0_0


7.5
5.0
2.5
Cover (1qm) [%]
0.30
0.32
0.34
0.36
0.38
Water capacity [%]

Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude r > 0.7 Dormann et al. 2013 Ecography DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cwm.")) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#     ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

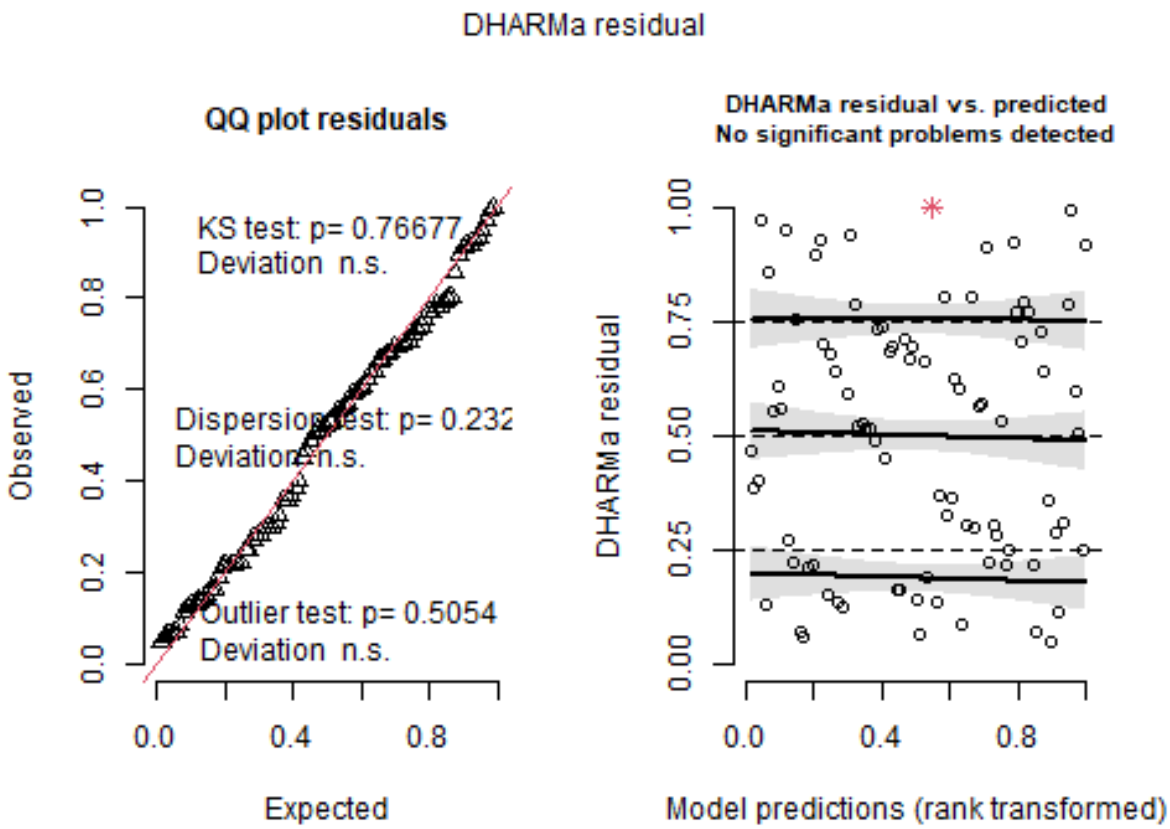
```
load(file = here("outputs", "models", "model_richness_timing_herbicide_nws_5.Rdata"))
#load(file = here("outputs", "models", "model_sla_esy4_3.Rdata"))
m_1 <- m5
#m_2 <- m2
```

```
formula(m_1)
## y ~ treatment * year + water_cap
# m_2@call
```

Model check

DHARMa

```
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```




```
# simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```

```
# plotResiduals(simulation_output_1$scaledResiduals, sites$treatment)
# plotResiduals(simulation_output_2$scaledResiduals, sites$treatment)
# plotResiduals(simulation_output_1$scaledResiduals, sites$year)
# plotResiduals(simulation_output_2$scaledResiduals, sites$year)
# plotResiduals(simulation_output_1$scaledResiduals, sites$water_cap)
# plotResiduals(simulation_output_2$scaledResiduals, sites$water_cap)
```

Check collinearity part 2 (Step 5)

Remove VIF > 3 or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
## there are higher-order terms (interactions) in this model
## consider setting type = 'predictor'; see ?vif
```

```
##                      GVIF Df GVIF^(1/(2*Df))
## treatment       68.429171  3        2.022430
## year            49.296296  3        1.914854
## water_cap        1.276823  1        1.129966
```

```
## treatment:year 1434.074074  9        1.497494
```

```r
# car::vif(m_2)
```

Model comparison

R2 values

```r
MuMIn::r.squaredGLMM(m_1)
##             R2m       R2c
## [1,] 0.3191247 0.3191247
# MuMIn::r.squaredGLMM(m_2)
```

AICc

Use AICc and not AIC since ratio n/K < 40 Burnahm & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```r
# MuMIn::AICc(m_1, m_2) %>%
#   arrange(AICc)
```

Predicted values

Summary table

```r
car::Anova(m_1, type = 3)
```

```
## Anova Table (Type III tests)
##
## Response: y
##                 Sum Sq Df F value Pr(>F)
## (Intercept)      0.461  1  0.2285 0.6341
## treatment       11.035  3  1.8236 0.1506
## year            12.458  3  2.0589 0.1134
## water_cap        2.392  1  1.1861 0.2798
## treatment:year  10.817  9  0.5959 0.7963
## Residuals      143.208 71
```

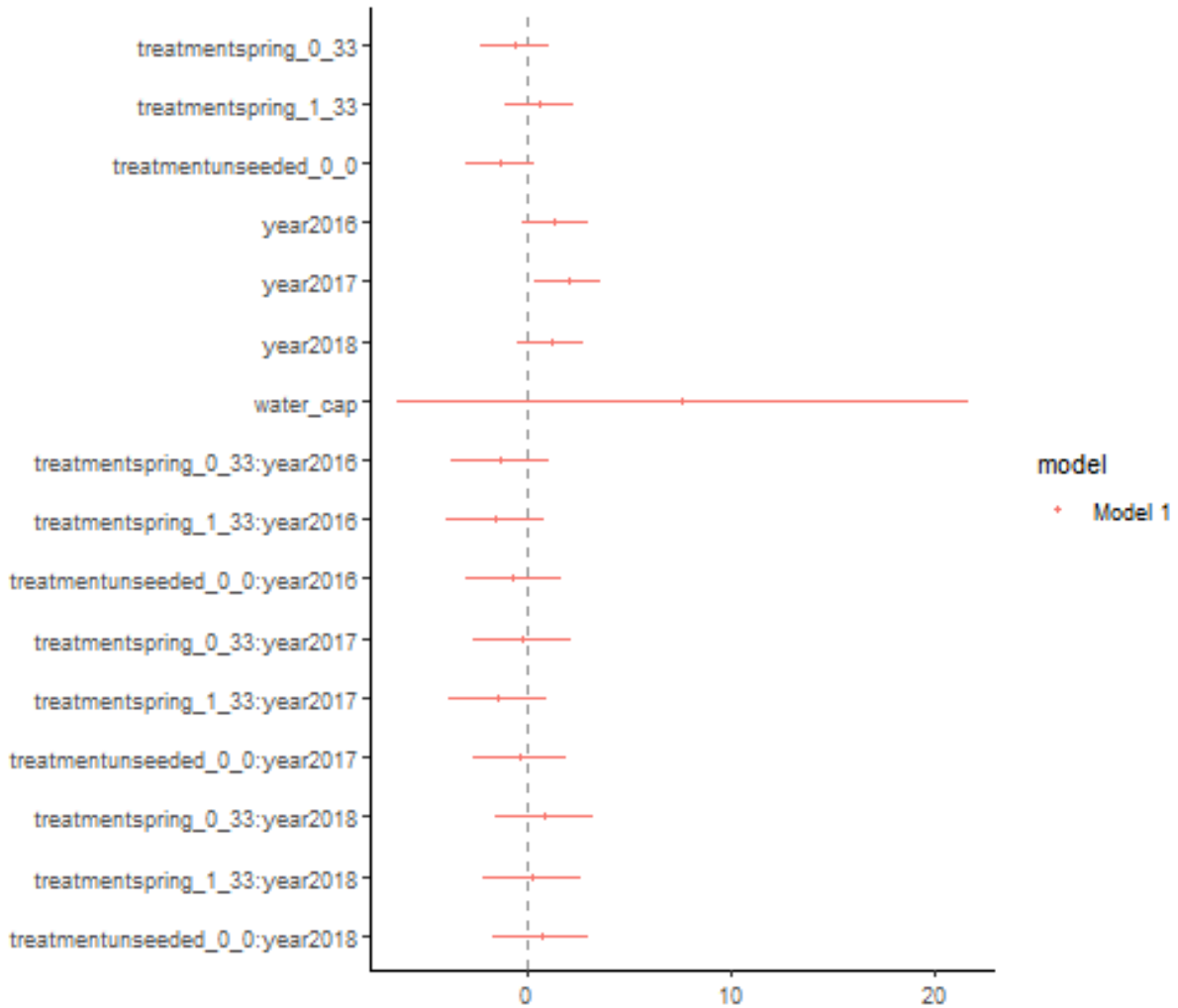
```r
summary(m_1)
```

```
##
## Call:
## lm(formula = y ~ treatment * year + water_cap, data = sites)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.6065 -0.9463  0.0236  0.8167  3.4801
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                      1.1422     2.3894   0.478   0.6341
## treatmentspring_0_33            -0.6117     0.8615  -0.710   0.4800
## treatmentspring_1_33             0.5899     0.8700   0.678   0.4999
## treatmentunseeded_0_0           -1.3114     0.8307  -1.579   0.1188
## year2016                         1.3333     0.8200   1.626   0.1084
## year2017                         2.0000     0.8200   2.439   0.0172 *
## year2018                         1.1667     0.8200   1.423   0.1592
```

```
## water_cap                        7.6720     7.0444   1.089   0.2798
## treatmentspring_0_33:year2016   -1.3333     1.2162  -1.096   0.2766
## treatmentspring_1_33:year2016   -1.5333     1.2162  -1.261   0.2115
## treatmentunseeded_0_0:year2016  -0.6667     1.1596  -0.575   0.5672
## treatmentspring_0_33:year2017   -0.2000     1.2162  -0.164   0.8698
## treatmentspring_1_33:year2017   -1.4000     1.2162  -1.151   0.2535
## treatmentunseeded_0_0:year2017  -0.3333     1.1596  -0.287   0.7746
## treatmentspring_0_33:year2018    0.8333     1.2162   0.685   0.4955
## treatmentspring_1_33:year2018    0.2333     1.2162   0.192   0.8484
## treatmentunseeded_0_0:year2018   0.6667     1.1596   0.575   0.5672
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.42 on 71 degrees of freedom
##   (8 Beobachtungen als fehlend gelöscht)
## Multiple R-squared:  0.3648, Adjusted R-squared:  0.2217
## F-statistic: 2.549 on 16 and 71 DF,  p-value: 0.003711
```

Forest plot

```r
dotwhisker::dwplot(
  list(m_1),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  theme_classic()
```



Effect sizes

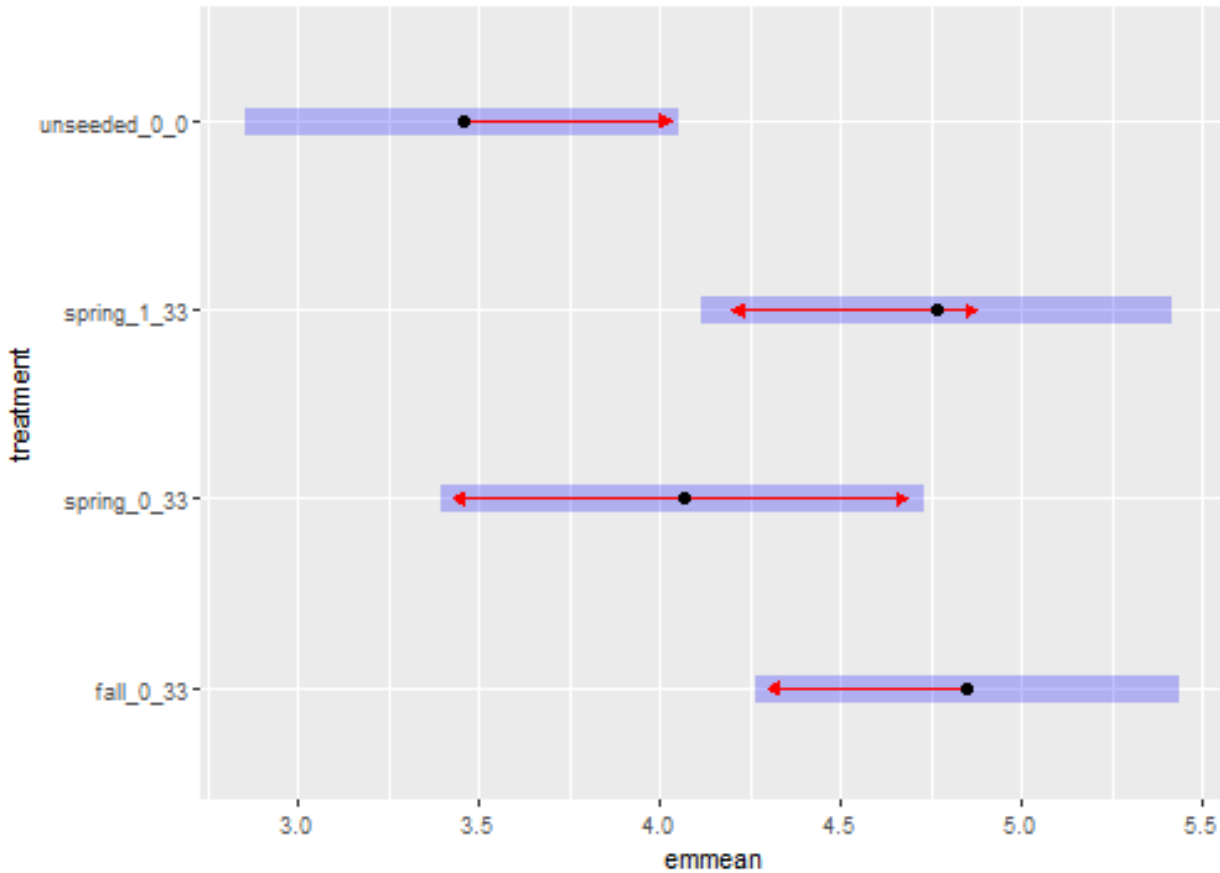
Effect sizes of chosen model just to get exact values of means etc. if necessary.

```
(emm <- emmeans(
  m_1,
  revpairwise ~ treatment,
  type = "response"
  ))
```

```
## $emmeans
##  treatment    emmean    SE df lower.CL upper.CL
##  fall_0_33      4.85 0.295 71     4.26     5.44
##  spring_0_33    4.06 0.335 71     3.40     4.73
##  spring_1_33    4.77 0.327 71     4.11     5.42
##  unseeded_0_0   3.46 0.300 71     2.86     4.06
##
## Results are averaged over the levels of: year
## Confidence level used: 0.95
##
```

```
## $contrasts
##  contrast                   estimate    SE df t.ratio p.value
##  spring_0_33 - fall_0_33     -0.7867 0.433 71  -1.817  0.2740
##  spring_1_33 - fall_0_33     -0.0851 0.450 71  -0.189  0.9976
##  spring_1_33 - spring_0_33    0.7016 0.485 71   1.448  0.4744
##  unseeded_0_0 - fall_0_33    -1.3947 0.431 71  -3.236  0.0097
##  unseeded_0_0 - spring_0_33  -0.6081 0.467 71  -1.301  0.5655
##  unseeded_0_0 - spring_1_33  -1.3097 0.430 71  -3.046  0.0168
##
## Results are averaged over the levels of: year
## P value adjustment: tukey method for comparing a family of 4 estimates
```

```r
plot(emm, comparison = TRUE)
```



Session info

```
## R version 4.5.0 (2025-04-11 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
```

```
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_1.11.1   DHARMa_0.4.7     glmmTMB_1.1.12   lme4_1.1-37
##  [5] Matrix_1.7-3     patchwork_1.3.1  ggbeeswarm_0.7.2 lubridate_1.9.4
##  [9] forcats_1.0.0    stringr_1.5.1    dplyr_1.1.4      purrr_1.1.0
## [13] readr_2.1.5      tidyr_1.3.1      tibble_3.3.0     ggplot2_3.5.2
## [17] tidyverse_2.0.0  here_1.0.1
##
## loaded via a namespace (and not attached):
##  [1] Rdpack_2.6.4           gridExtra_2.3          sandwich_3.1-1
##  [4] rlang_1.1.6            magrittr_2.0.3         compiler_4.5.0
##  [7] mgcv_1.9-3             vctrs_0.6.5            pkgconfig_2.0.3
## [10] crayon_1.5.3           fastmap_1.2.0          backports_1.5.0
## [13] labeling_0.4.3         utf8_1.2.6             ggstance_0.3.7
## [16] promises_1.3.3         rmarkdown_2.29         tzdb_0.5.0
## [19] nloptr_2.2.1           bit_4.6.0              xfun_0.52
## [22] later_1.4.2            parallel_4.5.0         R6_2.6.1
## [25] gap.datasets_0.0.6     stringi_1.8.7          qgam_2.0.0
## [28] RColorBrewer_1.1-3     car_3.1-3              boot_1.3-31
## [31] numDeriv_2016.8-1.1    estimability_1.5.1     Rcpp_1.1.0
## [34] iterators_1.0.14       knitr_1.50             zoo_1.8-14
## [37] parameters_0.27.0      httpuv_1.6.16          splines_4.5.0
## [40] timechange_0.3.0       tidyselect_1.2.1       rstudioapi_0.17.1
## [43] abind_1.4-8            yaml_2.3.10            MuMIn_1.48.11
## [46] doParallel_1.0.17      TMB_1.9.17             codetools_0.2-20
## [49] lattice_0.22-7         plyr_1.8.9             shiny_1.11.1
## [52] withr_3.0.2            bayestestR_0.16.1      coda_0.19-4.1
## [55] evaluate_1.0.4         marginaleffects_0.28.0 pillar_1.11.0
## [58] gap_1.6                carData_3.0-5          foreach_1.5.2
## [61] stats4_4.5.0           reformulas_0.4.1       insight_1.3.1
## [64] generics_0.1.4         vroom_1.6.5            rprojroot_2.0.4
## [67] hms_1.1.3              scales_1.4.0           minqa_1.2.8
## [70] xtable_1.8-4           glue_1.8.0             tools_4.5.0
## [73] data.table_1.17.8      mvtnorm_1.3-3          grid_4.5.0
## [76] rbibutils_2.3          datawizard_1.1.0       nlme_3.1-168
## [79] Rmisc_1.5.1            performance_0.15.0     beeswarm_0.4.0
## [82] vipor_0.4.7            Formula_1.2-5          cli_3.6.5
## [85] gtable_0.3.6           digest_0.6.37          farver_2.1.2
## [88] htmltools_0.5.8.1      lifecycle_1.0.4        mime_0.13
## [91] bit64_4.6.0-1          dotwhisker_0.8.4       MASS_7.3-65
```